Berna2

V 2.0.1



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In loving memory of my mother

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Between the 1950s and the mid 1960s, long before Robert Moog and Wendy Carlos pushed electronics into pop-music (with a few exceptions like the Barrons and Raymond Scott), electroacoustic music was pioneered by european radio laboratories, US universities. and tape centers. Composing with tapes and electronics was a serious painstaking and expensive affair, prerogative of a restricted elite of contemporary music composers and adventurous sound engineers. At that time there wasn't any electronic musical instruments market, as a matter of fact, most of the equipment was adapted from scientific tools belonging to radio engineering departments.
Sometimes the equipment was built from scratch cannibalizing anything that had wires, tubes and pots, more rarely, the studios used the few commercial instruments available in those days, such as the Melchord, the Trautonium and the Theremin.
Contrarily to what happens today, electronic music then was everything but fast and easy to create. A few minutes of electronic composition could take months of work. Everything was handmade, from complex timbres with multiple sine oscillators bounces to tape editing with scissors and scotch-tape. Even sound envelopes were manually made by cutting tapes' edgdes at different degrees of inclination. Ussachevsky's ADSR was yet to be invented!

Berna is a software simulation of a late 1950s electroacoustic music studio, mostly inspired by the RAI studio di Fonologia Musicale of Milan. Oscillators, filters, modulators, tape recorders, mixers, are all packed in a easy-to-use interface with historical accuracy.

Explore serial, concrete and tape music or create strange new sonic worlds with instruments inspired by the greatest studios of the early days of electronic music.
Are you ready to meet the grandfather of the synthesizer?

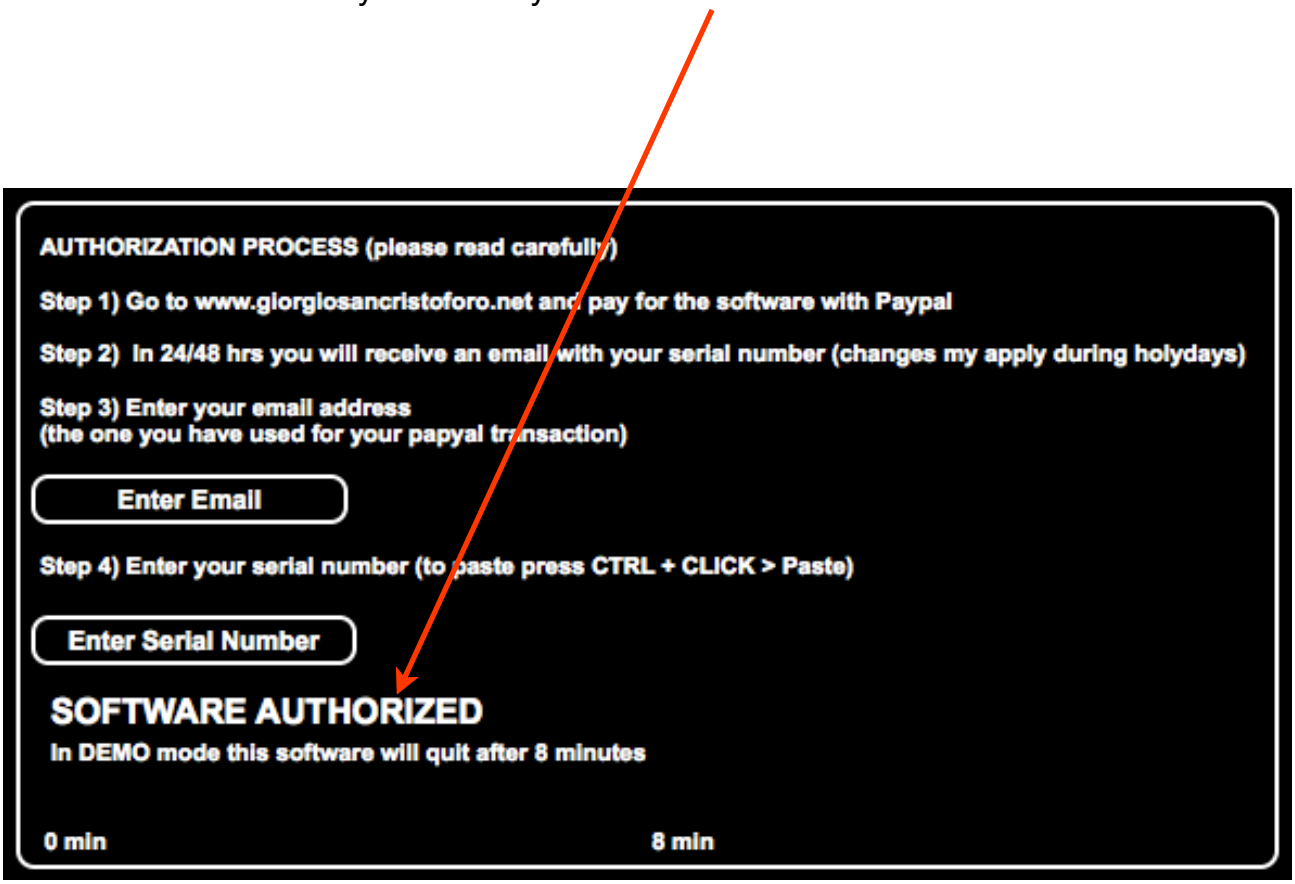
Berna 2.0, not only has a new **stunning realistic UI** but also it includes **new processors** like the Comparator, an 18 bands 1/3 Octave FilterBank and an 18 Bands Vocoder, plus new Tone Burst Generator, two Dynamic Modulators, new Reverb, new Frequency Shifter (with modulator input), new FM Oscillator (with modulator input) and 8 new Multi Waveform Oscillators!

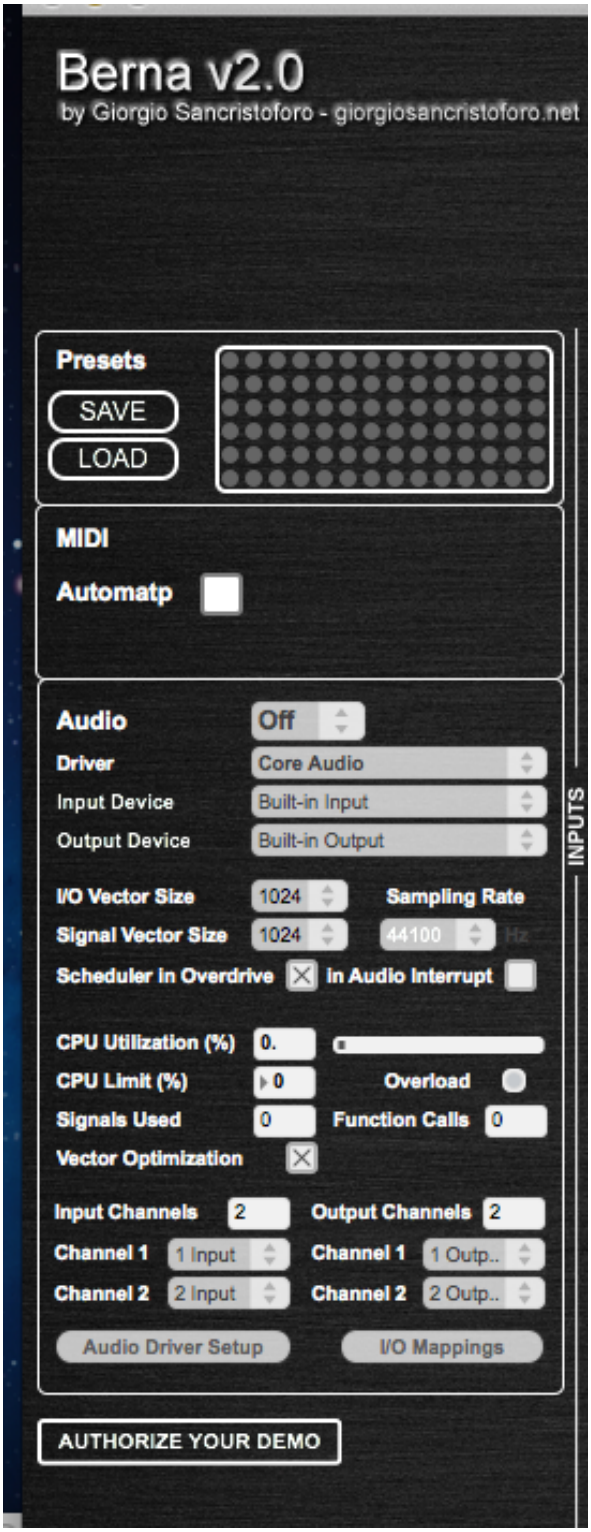
AUTHORIZATION

In DEMO MODE the software will run for 8 minutes, then it will quit.

1) Donwload form the Berna webpage the latest installer
2) On the same page, click BUY NOW button, you will be guided to Paypal
   (I can only accept paypal, please do not ask for other methods)
3) In 24/48hrs you should receive a mail containing your serial number (changes my apply during holydays, please be patient), the mail will be sent to the address you use for your paypal account.
4) Click MATRIX from the upper right menu and access the Authorization popup
   by clicking the AUTHORIZE YOUR DEMO button on the lower left of the page.
5) Input your email (your paypal email)
6) Input your serial as is with spaces (to use paste press CTRL + mouse click in the form and choose PASTE function.

If the software is correctly authorized you will see this





SETTING UP

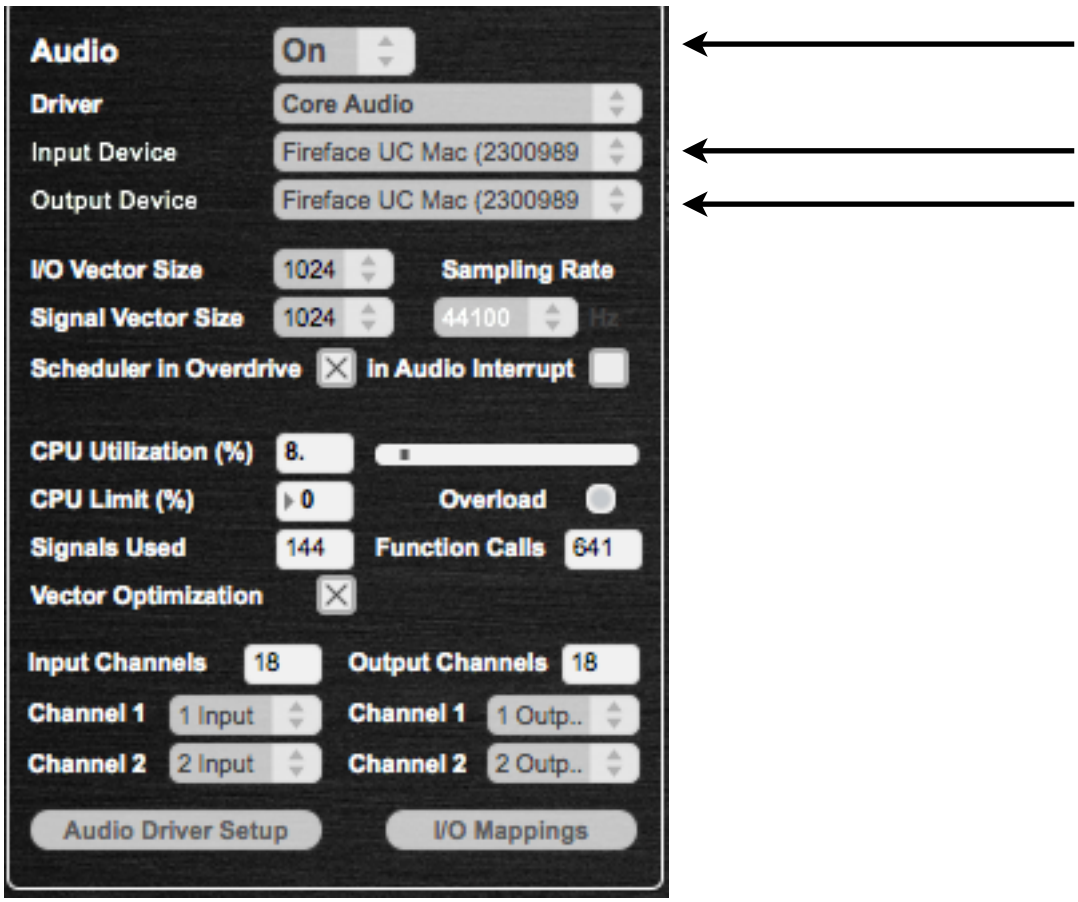
In the matrix page you will find three important features:

Presets:
save and load settings (excluding audio files)

MIDI:
Automap for your MIDI controller

Audio:
Setting for your audio interface

Before playing with Berna, please choose your audio interface I/O
and then **turn the DSP processor ON**



The sampling rate can be adjusted to your needs
Scheduler in Overdrive should be always ON
in Audio Interrupt should be always OFF

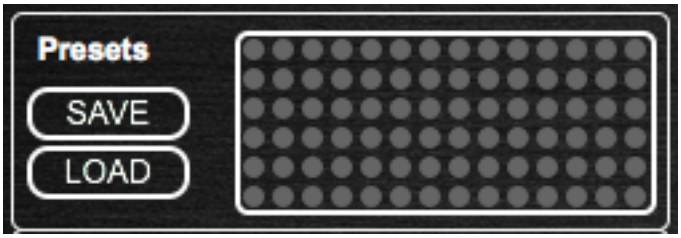
I/O Vector Size and Signal Vector Size

“The I/O Vector Size (I/O stands for input/output) con- trols the number of samples that are transferred to and from the audio interface at one time. The Signal Vector Size sets the number of samples that are calculated by MSP objects at one time. This can be less than or equal to the I/O Vector Size, but not more. The I/O Vector Size may have an effect on latency and overall performance. A smaller vector size may reduce the inherent delay betwe- en audio input and audio output, because MSP has to perform calculations for a smaller chunk of time. On the other hand, there is an additional computational burden each time MSP prepares to calculate another vector (the next chunk of audio), so it is easier overall for the
processor to compute a larger vector. However, there is another side to this story. When MSP calculates a vector of audio, it does so in what is known as an interrupt. If MSP is running on your computer, whatever you happen to be doing (word processing, for example) is interrupted and an I/O vector's worth of audio is calculated and played. Then the computer returns to its normally scheduled program. If the vector size is large enough, the computer may get a bit behind and the audio output may start to click

because the processing took longer than the computer expected. Reducing the I/O Vector Size may solve this problem, or it may not. On the other hand, if you try to generate too many interrupts, the computer will slow down trying to process them. Therefore, you'll

typically find the smaller I/O Vector Sizes consume a greater percentage of the computer's resources. Optimizing the performance of any particular signal network when you are close to the limit of your CPU's capability is a trial-and-error process. That's why MSP provides you with a choice of vector sizes."

Storing, recalling, saving and loading presets
Just like in Berna 1, in Berna 2 you can easily create and store prestes.



Presets can store any potentiometer, button, fader and matrix state and can be recalled very quickly.
NOTE: presets do not save audio files into the tape recorders. You will need to load them manually from the tape recorders.
To save a preset simply use shift+mouse click over one of the dots displayed in this interface.
To recall a preset just click the dot with the mouse.

To save the presets to a FILE, press SAVE
To load presets from a FILE, press LOAD

The software uses XML or JSON file format type.

MIDI
Berna 2 features a useful automap function for your MIDI controllers.
Connect your controller to the computer and press the Automap toggle



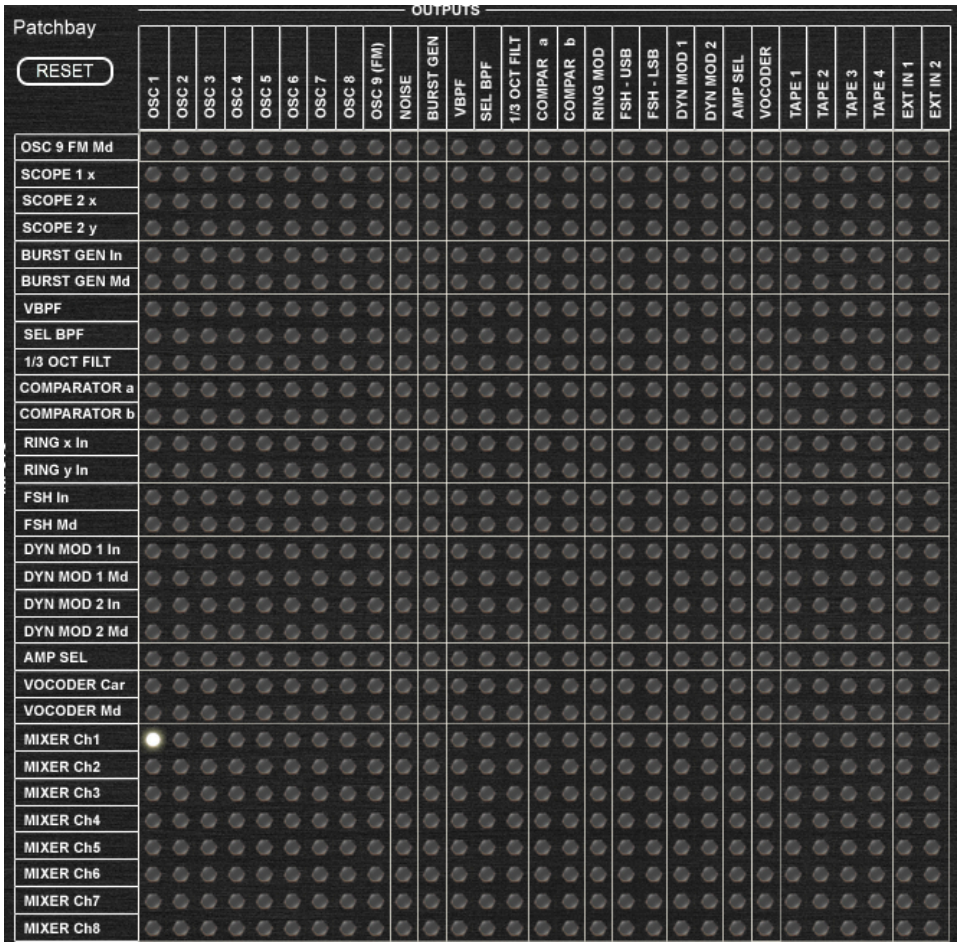
With Automap ON you will see a number of toggles appearing over the user interface.
To connect a Control Change to a parameter simply switch on the toggle on the desidered parameter, send some MIDI data from your controller (a fader or a pot). At this point you will see that your controller succesfully sends MIDI data to the desidered parameter.

Switch the toggle of the parameter to OFF and go on mapping other parameters with the same procedure. At the end switch OFF the Automap function.
When you will store a preset and save it to a file, all the automap work will be saved with the file.

THE USER INTERFACE

THE PATCHBAY

The patchbay is the routing facility of the studio, designed to give you total freedom on the signal flow without using messy patching cables. The use of the patchbay is straightforward: at the top you'll see the OUTPUTS and at the left you'll see the INPUTS of every module of the studio. Let's see a couple of examples.



OSCILLATOR 1 to the MIXER CH 1



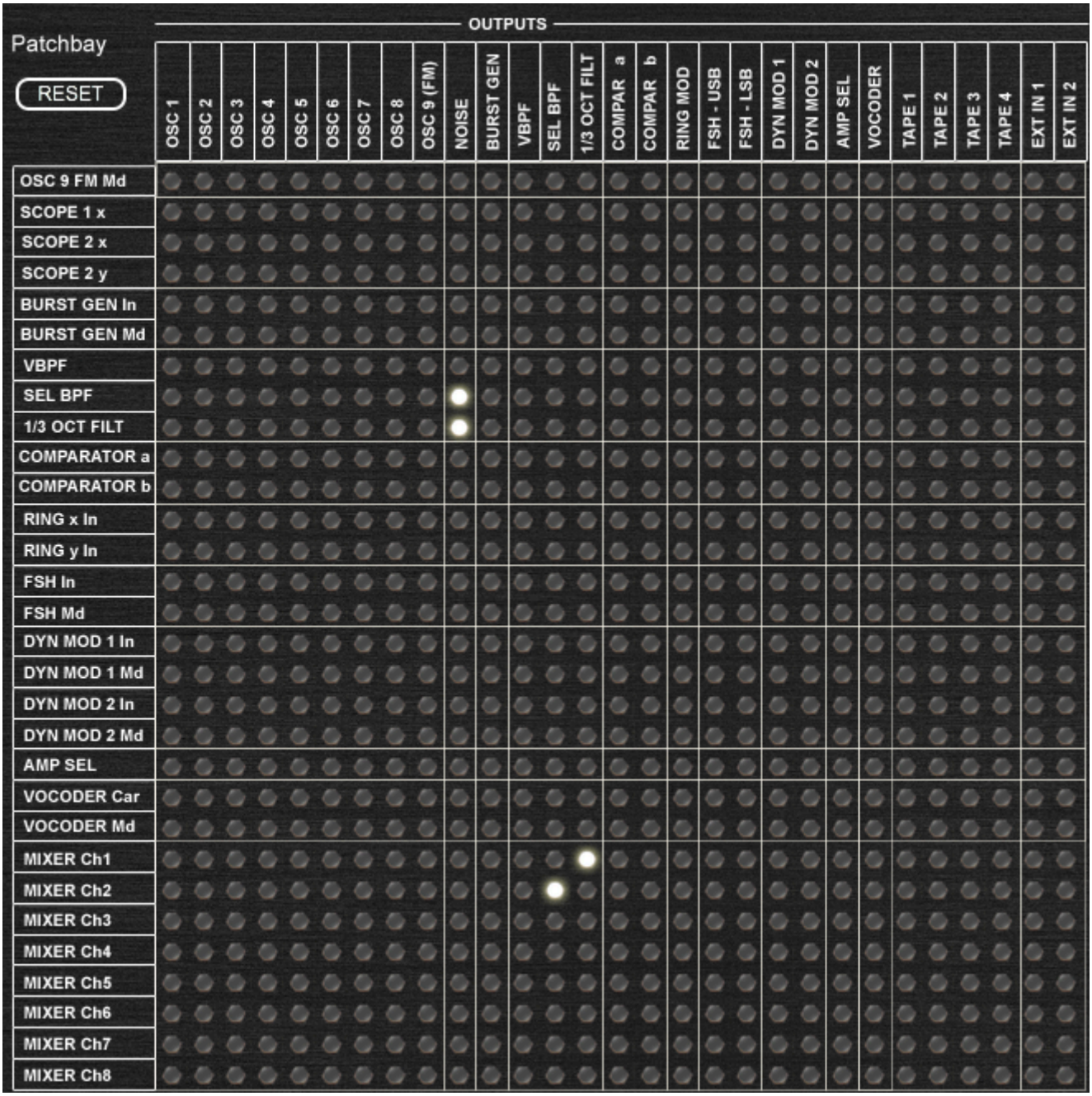
Rise OSCILLATOR 1's GAIN, select a frequency and a waveform



Now rise fader of CH1, you will hear the sound of the oscilator

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ANOTHER EXAMPLE:

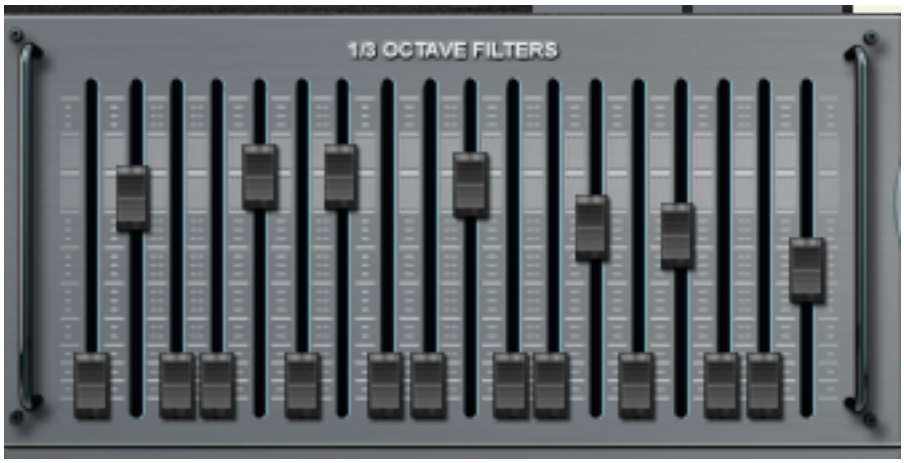


NOISE into 1/3 Filter and and Selective BPF, routed respectively to CH1 and CH2 of the MIXER

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Turn up noise volume and select noise type



Set frequency bands' amplitude on the 1/3 Oct Filter



Rise the gain of Selectove PBF, and select a frequency

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Rise faders 1 and 2 and PAN them if you like

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THE MODULES

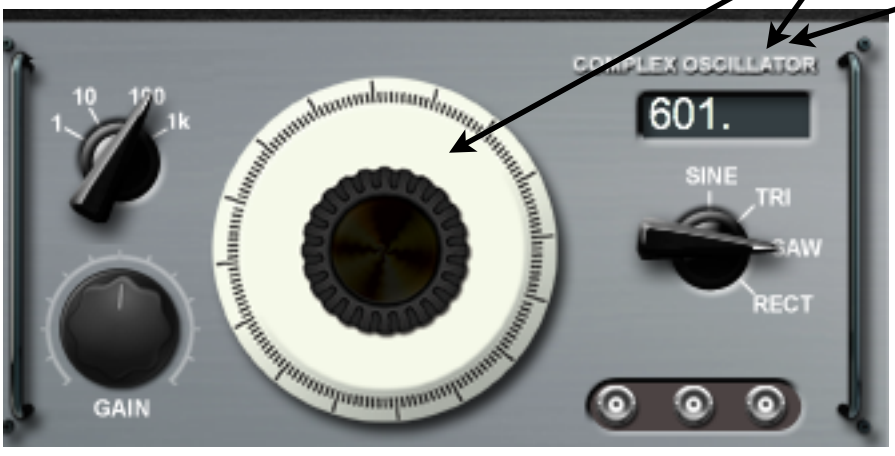
THE OSCILLATORS

The oscillators are numbered following this scheme



There are 8 multi-waveform oscillators and one sinusoidal oscillator with FM input

Each of the 8 multiwaveform oscillator is identical
They can produce waveforms ranging from 0.001 HZ to 1/2 of the sampling frequency
The Multi Waveform oscillator frequeny can be set with the mouse or with the keyboard (in the frequency display).



A selection of Sinusoid, triangle, sawthoot and rectangular waves is available, one at a time.

WITH MIDI
you can control GAIN and FREQUENCY

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The ninth oscillator has different features.
It is a Sinusoidal Oscillator with FM input (Osc9 FM Md in the patchbay). Try patching OSC 1 to OSC 9 FM Md with a sine at around 6Hz and a very low index of modulation. You'll get vibrato.
More radical modulations can be obtained at audio rate modulation.

Index controls the index of modulation.

! Remember to power it on !



The Noise generator outputs two different kind of noise:
White and Pink.
Overall amplitude is controllable with Gain

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The tone burst generator is a timed gate. It opens and close an audio gate accordignly to the frequency of a modulator signal.
Modulator signal must be a sinusoidal oscillator at full gain between 1 and 50 Hz

Send a sine (modulator) at full gain to Burst Gen Md,
then send another signal (carrier) to be gated to Burst Gen IN.

Patch Tone burst to a mixer channel and rise the channel fader.
Power ON the tone burst generator.

The Tone burst generator has two controls one for the closed gate timing and the other of the open gate timing. The tone burst will open and close it's gate multiplying the cycle duration of the modulator input by a factor of 2, 4, 8, 16, 32, 64, or 128.

With cycle duration 2 at the left and 4 at the right, and a modulator = 1HZ the tone burst will stay closed for 2 seconds, and open for 4 seconds.

The slope + or - will select positive or negative peaks of the modulator signal to clock the gate.
A switch will select gated or direct signal at the output.

! Remember to power on the Tone Burst Generator in order to make it work !

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THE OSCILLOSCOPES



There are two oscilloscope inside Berna 2.

The upper oscilloscope works with one channel
and serves to monitor waveforms

Try patching an oscillator to SCOPE 1X,
rise X Gain and adjust TIME/DIV (the internal clock of the oscilloscope)



The second oscilloscope uses 2 input (x and y) and it is useful to monitor harmonic relation between two signals.

Try patching oscillator 1 @ 400Hz to SCOPE 2 X
and oscillator 2 @ 801Hz to SCOPE 2 Y

A beat frequency of 1Hz will be visible in the Lissajous pattern displayed by the scope.
When the second oscillator is set @ 800 Hz (2 x frequency of oscillator 1) the pattern will be steady.

FREQUENCY SHIFTER

The frequency shifter will shift (up or down) the specturm of a signal accordignly to a modulator signal (sinusoid at full gain)

Turn on the Frequency Shifter.
Patch Oscillator 1 @ 1Hz into FSH Md (modulator)
Patch Oscillator 2 @ 100 Hz into FSH IN
Patch FSH USB (Upper Side Band) or FSH LSB (Lower Side Band) to mixer CH1
Rise CH1 fader

Now rise oscillator 1 frequency. Depending by USB or LSB output, the spectrum of the carrier signal will be transposed up or down.





THE AMPLITUDE SELECTOR

This device is an automatic (threshold dependent) signal gate.

When the input signal amplitude is higher than the threshold, the gate will open.

A control is given for fast or slow gate closing time

INs and OUTs are maked on the patchbay as AMP SEL

Fine adjustments of threshold can be obtained using mouse click + drag at the right of the decimal point

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THE DINAMYC MODULATOR



This module acts as an envelope follower.
It takes two signals, a carrier and a modulator.

The modulator envelope curve is extracted and applied to the carrier signal.

The Delay control delays the modulator signal

Balance controls the output mix, from dry to fully modulated.
Mod Index is the amplitude of the modulator signal
Control soften the transients.

TRY THIS:

Patch the Noise Generator to Amplitude Selector,
Turn up the Noise Generator gain
Patch the Amplitude Selector to Mixer CH 1, rise CH 1 Fader
Adjust the threshold until short burst of noise are heard

Upatch the amplitude selector from the mixer and patch it to DYNAMIC MODULATOR Md In (DYN MOD 1 Md)

Patch Oscillator 1 @ 1000Hz to DYN MOD 1 In

Patch DYN MOD 1 In to Mixer CH1

You will hear Oscillator 1 in short burts (similar to morse code),
the envelope coming out from the Amplitude Selector is now modulating the amplitude of the sine oscillator

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RING MODULATOR



A ring modulator with selectable full balanced modulation (ring) or qusi-balanced modulation.

INPUTS: RING X In
RING Y In

OUTPUTS: RING MOD

SELECTABLE BAND-PASS FILTER



This device includes a Low-Pass Filter and an High-Pass filter (2nd order) in serial configuration.

With Low-Pass fully open and High-Pass fully closed, the signal isn't affected.
Combinng the two filters, Low-Pass, Band Pass or High Pass filtering is possible.

SELECTIVE BPF



This is a special Band-Pass Filter with a very narrow band.
Try filtering white noise with this.
A flute-like sound will appear.

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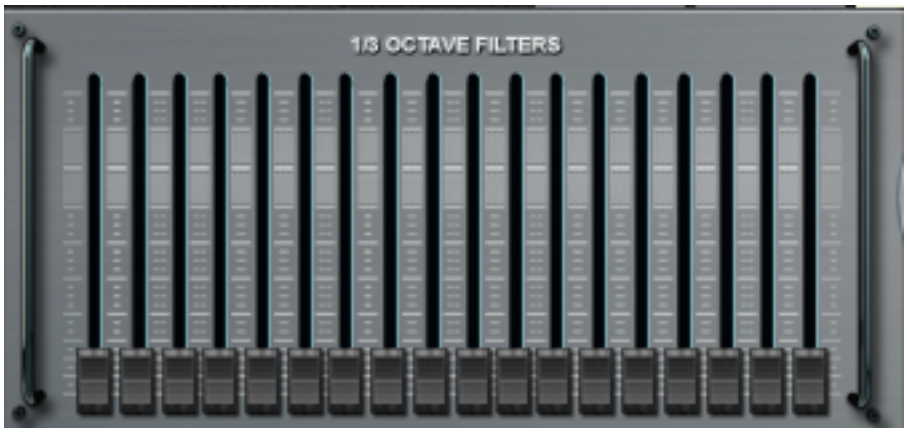
THE VOCODER



This is a 18 bands vocoder.
As every vocoder, it requires two signals to work.
A carrier (rich in harmonics like a sawthoot wave) and a modulator with transients (like a voice)

The spectral envelope of the modulator will be applied to the carrier signal.

1/3 OCTAVE FILTER



Not so much to say
This is a 18 bands band-pass filter.

THE COMPARATOR



A comparator takes two signals (A and B) and compare their amplitude.
If signal A is > B, a DC voltage will be sent to COMPAR a OUTPUT
If A < B a DC voltage will be sent to COMPAR b OUTPUT.
These outputs can only be used as modulator signal with the DYNAMIC MODULATOR
B can be internal (REF) or external (EXT).

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TAPE ECHO and PLATE REVERB



These two effects can be routed only from the mixer.
Tape Echo is on AUX 2, and Reverb on AUX 1

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MIXER

The mixer has 8 channes with PAN and 2 AUX (send [post-fader] and return)
The switch over the Master Fader route the Master to Tape 1, 2 3 or 4 Input

Tape returns have their output faders, a button select whether the tape return is routed to the Master Fader or to the Matrix (the patchbay) in order to process tape sounds with the modules.

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THE TAPE PLAYERS



There are 4 tape players

To IMPORT an Aiff or Wav file
press button [I]

To EXPORT an Aiff or Wav file
press button [E]

To record from the mixer, FIRST
select the leght of the tape (in
seconds)

Then press B

Now you are ready to record, press the record button.
Once recording is done press stop button

To play back press > play button
To play back in loop press loop button
When in LOOP, playback can be FFW or REV

TRIM is possible with button [T].
In order to trim a sound, select duration, then press T

Playback speed can be changed with this control

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